Part 5 presenter script: slides 34–41

Visualisation Cases and Defender Extension · Srinjoy Chakraborty · CS637, Stealthy Computational Delay Attacks on Control Systems (Nauta, Sandberg & Maggio, ICCPS 2025)

SLIDE	CONTENT	TIME
34	Divider	0:00–0:07
35	Furuta: three Table 1 HOLD attacks	0:07–0:50
Demo	Furuta dashboard, live	0:50–2:05
36	Furuta: horizon and detector validation	2:05–2:35
37	Tank: detector comparison	2:35–3:05
38	Tank: HOLD vs ZERO	3:05–3:35
39	Defender min–max formulation	3:35–4:10
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About 1,100 spoken words, timed for fast delivery at roughly 215 words a minute. Grey lines marked **[optional]** add about 25 seconds, so drop them if you are running long. Italic grey lines are stage cues and are not read out. Notation matches the edited deck (`deck/main.tex`): S_{χ^2} for the geometric statistic and $\chi^2_{.99}(4) = 13.277$ for its limit. Small grey brackets after a symbol give its meaning as a memory aid; say them aloud only if it helps the audience.

Before the talk: open `visualisations/furuta_table1_dashboard.html` in a browser tab (double-click it, or run `python main.py dashboard --system furuta`). Set it to *Paper Table 1: HOLD W=10*, Trajectory graph *Pendulum angle* and Playback *Normal*, then press ↻ **Reset**. Do **not** plan to demo `tank_verified_replay.html`. The copy in the repo is truncated partway through its bundled Plotly library, so it opens as a blank page.

Slide 34 · Visualisation Cases and Defender Extension

0:00–0:07

Gesture at the "Symbols used in this part" line under your name; you need not read it out.

Thank you, Abhishek. In this last part I'll replay the paper's attacks on both plants under both actuation policies, and then turn the question around to the defender.

Slide 35 · Furuta: the three published HOLD attacks

0:07–0:50

Point to the parameter line (it now notes that only Δ differs), then trace the curves.

These are the three Table 1 schedules for the Furuta pendulum: ℓ *(attack length in periods)* **= 25**, $\hat{p}_m$ *(nominal miss probability)* = 0.25, $\bar{a}$ *(watchdog limit on consecutive misses)* = 4, α *(significance level)* = 0.01, with a 5 ms period. They were optimised for **HOLD**, where a missed deadline reuses the previous command. We feed the same schedule and initial state into both policies. In Equation 4, only Δ *(actuator-memory block)* changes: identity for HOLD, zero for **ZERO**.

As the figure shows, the HOLD curves stay within a few degrees and then fall sharply in the last periods, while ZERO stays flat. In the table, window 10 uses fifteen misses and gives $\boldsymbol{\omega}$ *(extra terminal cost vs. no attack)* **= 0.144 rad²**, with a peak of almost **22°**. Windows 12 and 14 give 0.078 and 0.088. Recall from the Part 2 glossary that ω *(extra terminal cost vs. no attack)* is J_{attack} *(terminal cost under attack)* minus J_{nominal} *(terminal cost with no attack)*, the extra terminal cost from Equation 7. Under ZERO, ω *(extra terminal cost vs. no attack)* is about **minus 7×10^{-6}**, slightly

better than nominal. The intuition is that a held command keeps pushing in a direction that is no longer correct, while zero input just lets the pendulum coast.

Point to the right-hand box.

That does not bound ZERO's worst case, because the attacker was never re-optimised for it. Also, all three schedules pass the watchdog and binomial tests, but the geometric (χ^2) test of Equations 10 and 11 rejects them: S_{χ^2}*(geometric χ^2 statistic)* = **187, 86 and 95**, against a limit of $\boldsymbol{\chi^2_{.99}(4)}$*(critical value for $\bar{a}$ = 4)* = **13.277**. Let me show you why.

Live demo · Furuta dashboard

0:50–2:05

Switch to the browser tab. Every control named below is in the top control bar.

DO	SAY
1. Scenario *Paper Table 1: HOLD W=10*. Press ▶ **Play**.	This dashboard runs the authors' exact REP matrices: **HOLD** on the left, **ZERO** on the right, with the same schedule. The dashed line is nominal. The HOLD rod wobbles, then falls to 22°. The ZERO rod never exceeds its initial 0.57°.
2. Point to the red and teal strip, *Deadline-miss schedule*.	This strip is the attack. Red is a miss and teal is a hit. The pattern is bursts of three misses, then a hit. Three is under the watchdog limit of four.
3. Point to the *Binomial* chart.	This is the binomial test, Equation 9. Each bar is the miss count in one 10-period window, and the dashed line is the allowance of 6. Many bars touch it and none cross it, so the optimiser spends the whole budget.
4. Point to the *Geometric/χ^2* chart (geometric run-length test), then the **Detector audit** pills.	The geometric test asks how many misses came before each completed job. Normally that count is geometric, mostly zero. With 11 jobs we expect about 8 in category zero and 0.13 in category three. We observe 6 and **5**, which gives S_{χ^2}*(geometric χ^2 statistic)* = 187, far above $\chi^2_{.99}(4)$*(critical value for $\bar{a}$ = 4)* = 13.277. The audit shows watchdog and binomial passing and the geometric (χ^2) test failing, marked "not enforced" because this attack only optimised against the binomial test.
5. Set **Trajectory graph** to *Applied-control state*. Drag **Step** from 0 to 25.	This is the mechanism. Under HOLD the command is flat across each burst, the same stale value four periods in a row. Under ZERO it drops to zero. At the end, HOLD's held command pushes the wrong way just as the pendulum falls.
6. Set **Trajectory graph** to *Accumulated objective difference*.	For HOLD the impact stays near zero, then climbs steeply to 0.144 at the last sample. Only the terminal state is scored, so the attacker saves the damage for the end.
7. Switch **Attack scenario** to *HOLD W=12*, then *W=14* (Playback *Fast*).	The other rows behave the same way. Window 12 gives 0.078 and 16°, and window 14 gives 0.088 and 17°, so the trend is not monotone. Their S_{χ^2}*(geometric χ^2 statistic)* values of 86 and 95 also fail. Back to the slides.

Slide 36 · Furuta: horizon and detector validation

2:05–2:35

Point to the curves, the ωZERO(extra terminal cost under ZERO) column, then the "Extra, ℓ(attack length in periods) = 23" row.

These are the paper's ZERO-optimised schedules from Section 5.4.1, under the **combined detector** with window 20. As the figure shows, the angles stay below one degree. In the table, ω_{ZERO}*(extra terminal cost under ZERO)* rises from 2.74×10^{-5} at ℓ*(attack length in periods)* = 5 to 8.46×10^{-5} at ℓ*(attack length in periods)* = 15, then falls to 2.88×10^{-5} at 25. So **a longer horizon does not mean more terminal damage**. All three pass, with S_{χ^2}*(geometric χ^2 statistic)* of 6.33, 12.83 and 12.11.

[optional] 12.83 is just under 13.277, so the geometric (χ^2) constraint is binding.

The extra 23-period replay passes the binomial test only. Its S_{χ^2}*(geometric χ^2 statistic)* is 126, and under HOLD it reaches 4.2×10^{-2}, hundreds of times larger. Impact depends on the schedule and the active detector.

Slide 37 · Quadruple tank: detector comparison and replay

2:35–3:05

Point to the curves, then the first two table rows.

Now the quadruple tank, which is stable and non-minimum phase, with HOLD and T_s*(sampling period)* = 0.1 s. With ten periods we **enumerate all 1,024 schedules**, so these are exact optima. As the figure shows, output y_2*(lower-tank output)* deviates most under the binomial-only optimum: ω*(extra terminal cost vs. no attack)* = 0.003162, but S_{χ^2}*(geometric χ^2 statistic)* is **338.8**. Adding the geometric (χ^2) constraint gives 0.002074 with S_{χ^2}*(geometric χ^2 statistic)* = 10.71, which passes. That is a **34.4% reduction**, consistent with the paper's Figure 3: the run-length test removes the bursty, most damaging schedules.

[optional] This compares constraint sets, not detectors at equal false-alarm rates, and the curves are deviations, not absolute water heights.

Slide 38 · Quadruple tank: policy comparison

3:05–3:35

Point to the paired bars, then the equation and the small legend under it.

Our extension applies ZERO to the tank by changing only the actuator-memory block to zero. That means zero *input deviation*, not pumps off. As the bar chart shows, all three schedules do **more** damage under ZERO: **2.92×, 2.12× and 2.48×**. So the pendulum lesson reverses. The pendulum is unstable and fast, so a stale command quickly becomes harmful. The tank is slow, so the held command is still roughly correct, and dropping it removes the correction. On units, the tank output is $y_k = \tfrac{1}{2}\,[x_{1,k}, x_{2,k}]^\top$ *(y_k = tank output at step k; $x_{1,k}$, $x_{2,k}$ = water-level deviations of the two lower tanks)*, so with a quadratic cost $\omega_y = \tfrac{1}{4}\,\omega_{REP}$ *(ω_y = extra terminal cost measured on the output; ω_{REP} = the same cost computed on the raw states in the authors' REP code)*. Ranking policies would need the attacker re-optimised under each.

Slide 39 · Defender study: formulation and status

3:35–4:10

Trace the equation from the inside out; the legend under it defines every symbol.

Now the defender. As the equation shows, $d^* = \arg\min_d \max_{\ell \in L} \omega^*(d, \hat{p}_m, \ell)$ *(d^* = best defender configuration; d = a defender configuration (policy, detector, window, α); ℓ = attack length; L = set of tested attack lengths; ω^* = attacker's best extra cost; $\hat{p}_m$ = nominal miss probability)*, subject to $FA(d, \hat{p}_m) \le \varphi$ *(FA = session false-alarm rate; φ = allowed false-alarm budget)*. Reading it from the inside out: ω^**(attacker's best extra cost)* is the attacker's best stealthy schedule from Equation 7, the max takes the worst horizon, and the min picks the policy, detector, window and α*(significance level)* that make that worst case smallest. The false-alarm constraint matters, because without it the defender would just pick the tightest detector and alarm on normal jitter. FA*(false-alarm rate)* is a *session* rate, estimated by simulating nominal Bernoulli sessions. α*(significance level)* is only per-test, and $\hat{p}_m$*(nominal miss probability)* is measured, not chosen. In the

status table, the replays are done, the binomial sweep is partial, the geometric (χ^2) and combined sweeps are planned, and ℓ*(attack length in periods)* = 25 is missing.

[optional] The binomial grid is 36 choices per scenario: 540 quick-grid runs, or 2,700 for the full grid.

Slide 40 · Defender validation: horizon coverage

4:10–4:40

Trace the top curve, then point to the table and 6.56.

As the figure shows, for $\hat{p}_m$*(nominal miss probability)* = 0.25 the maximum peaks at window 10 at 0.022, dips at 12, rises at 14, and comes back up at 20, so it is **not monotone**. Lower $\hat{p}_m$*(nominal miss probability)* curves sit far lower, because a stricter nominal profile gives the attacker less cover. The table is the punchline. The saved maximum is 0.021973, but the ℓ*(attack length in periods)* = 25 Table 1 attack we just replayed reaches 0.144, which is **6.56 times larger**. Those solves failed on Gurobi error 10010, a licence size limit, so only 432 of 540 runs were saved. **A feasible attack is a lower bound. A partial maximum is never a safety guarantee.**

Slide 41 · Conclusion

4:40–5:05

Point to the left column, the right column, then Limitations.

To conclude, the paper shows that deadline misses alone, with no forged data, are an attack surface, and that impact depends on the plant, the schedule and the policy. We independently enumerated the $\bar{a}$*(watchdog limit on consecutive misses)* = 3 curve, showed that all three Table 1 schedules fail the geometric (χ^2) test, found that ZERO helps the pendulum but hurts the tank, and formulated a min–max defender study. Our limitations are the terminal-cost objective, $\hat{p}_m$*(nominal miss probability)* $\leq$ 0.25, no hardware, and the missing ℓ*(attack length in periods)* = 25 and false-alarm runs. The takeaway is that timing, policy and detection interact, so no single one picks the defence. Thank you.

Backup: quick answers for Q&A

not timed

Why does the geometric (χ^2) test reject when binomial passes?	The binomial test only counts misses per window. The geometric test checks *how they are grouped*. Bursts of exactly 3 are very rare with independent misses at $\hat{p}$ *(nominal miss probability)* = 0.25: about 0.13 such runs expected, 5 observed.
Where does 13.277 come from?	It is $\chi^2_{0.99}$ *(99% critical value of the χ^2 distribution)* with 4 degrees of freedom ($\bar{a}$ *(watchdog limit on consecutive misses)* = 4, α *(significance level)* = 0.01). REP implements the scaled form T *(scaled statistic)* = N *(number of completed jobs)* $\cdot S$ *(χ^2 statistic)* $\leq$ critical$\cdot N$ so that it stays tractable in the solver.
Is ZERO safer?	Not universally. It helps the unstable pendulum and hurts the stable tank under the same schedules. Neither is a worst case, because the attacker was not re-optimised for ZERO.
Is 34.4% a detector ranking?	No. It compares constraint sets at the same α *(significance level)*. A fair ranking needs equal session false-alarm rates.
Why is ω *(extra terminal cost vs. no attack)* **negative?**	ω *(extra terminal cost vs. no attack)* = J_{attack} *(terminal cost under attack)* − $J_{nominal}$ *(terminal cost with no attack)*. A negative ω *(extra terminal cost vs. no attack)* means that schedule left a smaller terminal angle cost than the nominal run. Only the final state is scored.
Why is the ℓ *(attack length in periods)* **= 25 data missing?**	Gurobi error 10010: the model exceeded the size-limited licence. It was not a timeout. 108 of 540 runs failed.

> Sources checked: `presentations/final/main.pdf` pp. 34–41 and their speaker notes in `main.tex`; `data/ppt/case_results.csv`; `data/ppt/defender_available_summary.csv`; the data embedded in `visualisations/furuta_table1_dashboard.html` (expected run-length counts, actuator traces); `docs/PPT_FILE_MAP.md`.